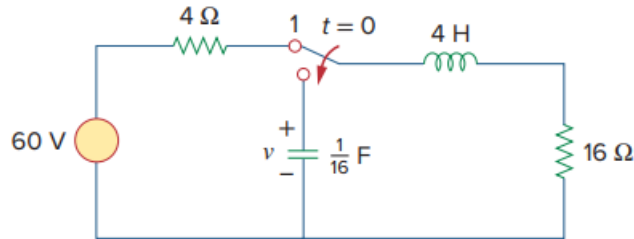


Problem

The switch in Fig. 8.105 has been in position 1 for $t < 0$. At $t = 0$, it is moved from position 1 to the top of the capacitor at $t = 0$. Please note that the switch is a make before break switch, it stays in contact with position 1 until it makes contact with the top of the capacitor and then breaks the contact at position 1. Given that the initial voltage across the capacitor is equal to zero, determine $v(t)$.

Figure 8.105:



Step-by-step solution

Step 1 of 11

Refer to Figure 8.105 in the text book.

At $t < 0$, inductor acts as short circuit.

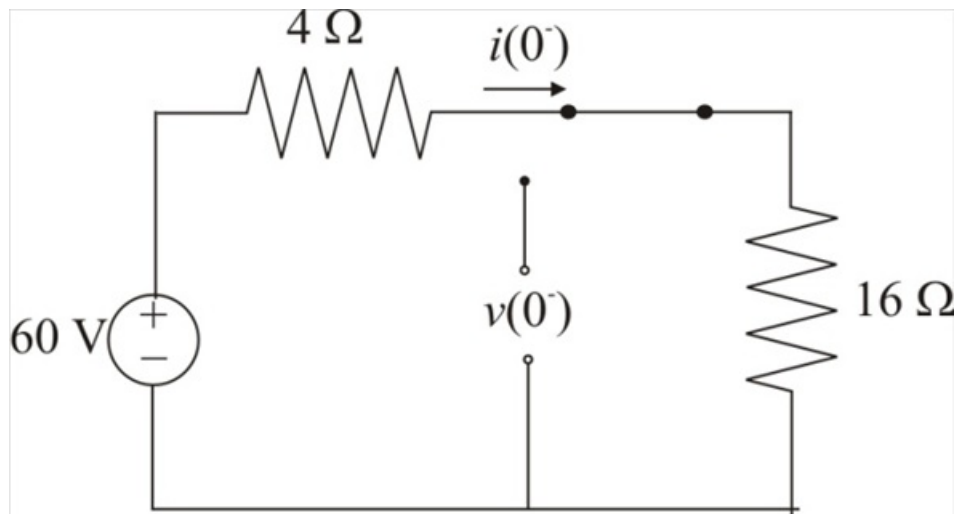


Figure 1

The value of the voltage $v(0^-)$ is 0 V.

Step 2 of 11

Calculate the value of the current $i(0^-)$.

$$\begin{aligned} i(0^-) &= \frac{60}{16 + 4} \\ &= \frac{60}{20} \\ &= 3 \text{ A} \end{aligned}$$

Therefore, the value of the current $i(0^-)$ is 3 A.

Draw the circuit diagram at $t > 0$.

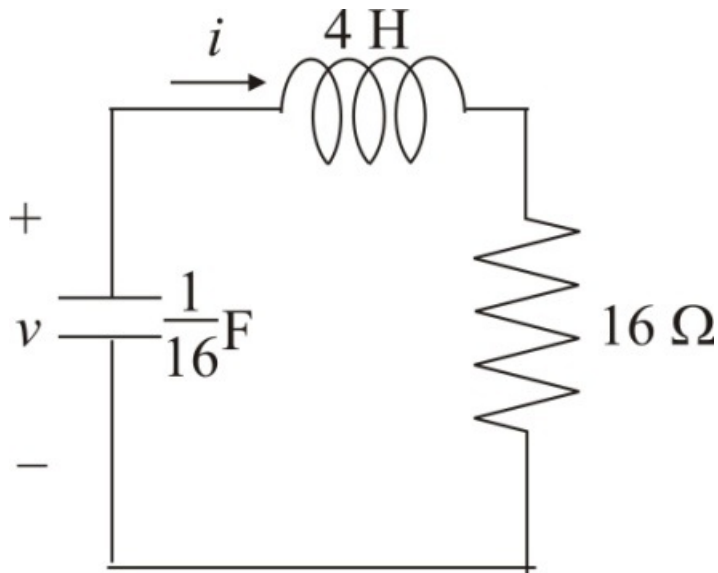


Figure 2

Step 4 of 11

The voltage across the capacitor and the current through the inductor will not change instantaneously. Hence, at $t > 0$ the value of the voltage across the capacitor $v(0^+)$ is 0 V and the value of the current $i(0^+)$ is 3 A.

Notice that the voltage across the capacitor and the current through the inductor are continuous. Therefore, the values at $t = 0$ are $v(0) = 0$ V and $i(0) = 3$ A.

Step 5 of 11

Consider the expression for the current through the capacitor.

$$\frac{1}{16} \frac{dv}{dt} = -i$$

$$\frac{dv}{dt} = -16i$$

Calculate the value of $\frac{dv}{dt}$ at $t = 0$.

$$\frac{dv(0)}{dt} = -16i(0)$$

Substitute 3 A for $i(0)$.

$$\begin{aligned} \frac{dv(0)}{dt} &= -16(3) \\ &= -48 \text{ V/s} \end{aligned}$$

Therefore, the value of $\frac{dv(0)}{dt}$ is -48 V/s.

Step 6 of 11

Apply Kirchhoff's Voltage Law in the loop.

$$4 \frac{di}{dt} + 16i + \frac{1}{(1/16)} \int i dt = 0$$

$$4 \frac{di}{dt} + 16i + 16 \int i dt = 0$$

Substitute $\frac{1}{16} \frac{dv}{dt}$ for i .

$$4 \frac{d}{dt} \left(\frac{1}{16} \frac{dv}{dt} \right) + 16 \left(\frac{1}{16} \frac{dv}{dt} \right) + 16 \int \left(\frac{1}{16} \frac{dv}{dt} \right) dt = 0$$

$$\frac{1}{4} \frac{d^2 v}{dt^2} + \frac{dv}{dt} + v = 0$$

$$\frac{d^2 v}{dt^2} + 4 \frac{dv}{dt} + 4v = 0$$

Treat $\frac{di}{dt}$ as s .

$$s^2 v + 4s v + 4v = 0$$

$$(s^2 + 4s + 4)v = 0$$

Step 7 of 11

Consider the characteristic equation of the circuit.

$$s^2 + 4s + 4 = 0$$

$$(s + 2)(s + 2) = 0$$

Therefore, the roots of the equation are -2 and -2 .

The circuit exhibits critical damped response since the roots of the characteristic equation are negative and equal.

Step 8 of 11

Draw the circuit for $t = \infty$.

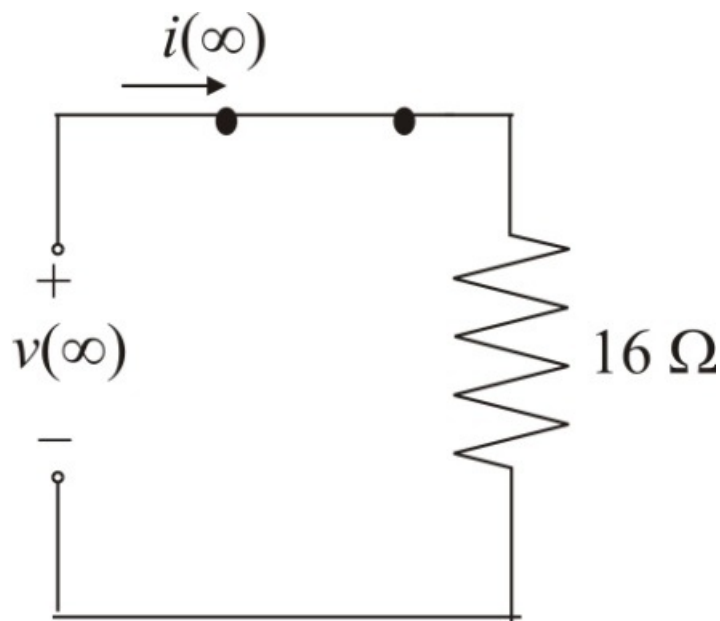


Figure 3

The value of the current $i(\infty)$ is 0 A and the value of the voltage $v(\infty)$ is 0 V .

Write the general solution for the voltage $v(t)$ for under-damped condition.

$$v(t) = v(\infty) + (A_1 + A_2 t) e^{\alpha t}$$

Substitute 0 V for $v(\infty)$ and -2 for α .

$$v(t) = 0 + (A_1 + A_2 t) e^{-2t}$$

$$v(t) = (A_1 + A_2 t) e^{-2t} \dots\dots (1)$$

Substitute 0 for t .

$$\begin{aligned} v(0) &= (A_1 + A_2(0)) e^{-2(0)} \\ &= (A_1 + 0) e^{(0)} \\ &= A_1(1) \\ &= A_1 \end{aligned}$$

Substitute 0 V for $v(0)$.

$$A_1 = 0$$

Therefore, the value of A_1 is 0.

Step 10 of 11

Recall equation (1).

$$v(t) = (A_1 + A_2 t) e^{-2t}$$

Apply derivative with respect to t .

$$\frac{dv(t)}{dt} = (A_1 + A_2 t)(-2) e^{-2t} + e^{-2t} A_2$$

Substitute 0 for t and 0 for A_1 .

$$\begin{aligned} \frac{dv(0)}{dt} &= (0 + A_2(0))(-2) e^{-2(0)} + e^{-2(0)} A_2 \\ &= (0 + 0)(-2) e^{(0)} + e^{(0)} A_2 \\ &= (0)(-2)(1) + (1) A_2 \\ &= 0 + A_2 \end{aligned}$$

Substitute -48 V/s for $\frac{dv(0)}{dt}$.

$$A_2 = -48$$

Therefore, the value of the constant A_2 is -48 .

Step 11 of 11

Recall equation (1).

$$v(t) = (A_1 + A_2 t) e^{-2t}$$

Substitute -48 for A_2 and 0 for A_1 .

$$\begin{aligned} v(t) &= (0 - 48t) e^{-2t} \\ &= -48te^{-2t} \text{ V} \end{aligned}$$

Therefore, the value of the voltage $v(t)$ is $\boxed{-48te^{-2t} \text{ V}}$.